

Aerospace Math Appendix — eVTOL Mini-Lab (Corrected Edition)

Notation uses SI units unless noted (L, D, T, W, ρ, V, S, C_L, C_D, AR, e, η_p, η_e).

Aerodynamics

$$q = \frac{1}{2} \rho V^2$$

Dynamic pressure q combines air density ρ and true airspeed V to set aerodynamic load scale (Pa).

$$L = \frac{1}{2} \rho V^2 S C_L$$

Lift L depends on dynamic pressure q, reference area S, and lift coefficient C_L (N).

$$D = \frac{1}{2} \rho V^2 S C_D$$

Drag D scales with dynamic pressure q, area S, and drag coefficient C_D (N).

$$C_D = C_{D0} + k C_L^2, \quad k = \frac{1}{\pi e AR}$$

Total drag coefficient equals parasite C_{D0} plus induced term with factor k set by Oswald efficiency e and aspect ratio AR (-).

$$V_S = \sqrt{\frac{2W}{\rho S C_{Lmax}}}$$

Stall speed V_S occurs when maximum lift coefficient C_{Lmax} can just support weight W (m/s).

$$\sin \gamma = \frac{T - D}{W}$$

Flight-path angle γ relates to excess thrust over drag (steady, small-angle climb).

$$n = \frac{L}{W}$$

Load factor n is the ratio of aerodynamic lift to weight and governs structural and maneuver limits (-).

Propulsion

$$P_{req} = D V$$

Propulsive power required equals aerodynamic drag D times airspeed V (W).

$$\eta_p = \frac{T V}{P_{shaft}}$$

Propeller efficiency η_p is useful power to the air T·V over shaft power into the propulsor (-).

$$P_{elec} = \frac{P_{air}}{\eta_p \eta_e}$$

Electrical bus power equals air power divided by propeller and electrical efficiencies η_p and η_e (W).

$$P_{hover} = \frac{W^{3/2}}{\sqrt{2\rho A}}$$

Ideal rotor hover power from momentum theory (induced only, neglects profile and figure-of-merit losses) (W).

$$T = 2 \rho A v_i^2, \quad P = T v_i$$

In hover, rotor thrust T and induced power P follow from induced velocity v_i through the disk (N, W).

$$E_{batt} = V_{nom} Q$$

Battery energy equals nominal voltage V_nom multiplied by capacity Q (Wh).

Performance

$$ROC = \frac{P_{avail} - P_{req}}{W}$$

Rate of climb equals power excess divided by weight (m/s).

$$R_{prop} \approx \frac{\eta_p}{c} \frac{L}{D} \ln \left(\frac{W_i}{W_f} \right)$$

Propeller-aircraft range scales with propulsive efficiency η_p , specific fuel consumption c (1/s), and lift-to-drag L/D (m).

$$R_{elec} \approx \frac{\eta_p \eta_e E_{batt}}{W} \frac{L}{D}$$

Approximate electric range assuming steady level flight and constant efficiencies (m).

$$t_{elec} = \frac{E_{batt}}{P_{elec}}$$

Electric endurance equals available battery energy divided by electrical power draw (s or h).

$$V_{(L/D)_{max}} : \frac{C_L}{C_D} \text{ max}$$

Best-glide speed occurs where lift-to-drag ratio C_L/C_D is maximized (m/s).

$$R = \frac{V^2}{g \tan \phi}$$

Coordinated-turn radius R depends on speed V and bank angle ϕ (m).

$$\omega = \frac{g \tan \phi}{V}$$

Coordinated-turn rate ω increases with bank ϕ and decreases with airspeed V (rad/s).

Navigation / Environment

$$\vec{V}_g = \vec{V}_a + \vec{W}$$

Groundspeed vector equals airspeed plus wind, forming the wind triangle for navigation (m/s).

$$\rho(h) \approx \rho_0 e^{-h/H}$$

Standard-day density decays approximately exponentially with geometric altitude h and scale height H (kg/m³).

$$TAS \approx IAS \sqrt{\frac{\rho_0}{\rho}}$$

True airspeed increases with altitude relative to indicated airspeed through density ratio (m/s).

$$FoS = \frac{E_{batt} - (E_{need} + E_{reserve})}{E_{batt}}$$

Energy factor-of-safety compares remaining battery energy to required plus reserve for mission planning (-).